

EduPulse AI — Independent V2 Validation Report

Model 3 Experiment B Generalization Check | Generated: August 13, 2026

1. Validation Setup & Data Specs

Attribute	Specification
Validation Sample Count	20,000 unseen records
Random Seed	2026 (different from V2 dataset)
Feature Count	20 behavioural metrics
Number of Classes	8 Target Classes
Loaded Model Artifact	models/recommendation/v2/best_model_v2.pkl
Model Type	Random Forest
Retraining Status	NO (0 fit calls executed)
Duplicate Rows vs V2 Data	0 (PASS - Genuinely Unseen)

2. Overall Validation Performance Metrics

Metric	Value
Accuracy	0.9782 (97.82%)
Weighted Precision	0.9789
Weighted Recall	0.9782
Macro Precision	0.9788
Macro Recall	0.9798
Weighted F1	0.9783
Macro F1	0.9790
Macro Specificity	0.9969
Weighted ROC AUC	0.9969
Matthews Corr Coef (MCC)	0.9751

3. Per-Class Performance Summary

Class	Class Name	Precision	Recall	F1	Support
0	Continue Current Skill	1.0000	0.9397	0.9689	2,685
1	Start Focus Session	0.9699	0.9858	0.9778	2,386
2	Take Short Break	0.9860	1.0000	0.9929	2,394
3	Practice Coding	0.9947	0.9838	0.9892	1,914
4	Revision	0.9841	0.9642	0.9741	3,662
5	Watch Learning Video	0.9990	0.9754	0.9871	2,151
6	Complete Pending Tasks	0.9554	1.0000	0.9772	2,484

7	Attempt Quiz	0.9415	0.9897	0.9650	2,324
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4. Generalization Comparison (Test vs Validation)

Metric	Original V2 Test	Independent Validation	Difference (Val - Test)
Accuracy	0.9781	0.9782	+0.0001
Weighted F1	0.9781	0.9783	+0.0001
Macro F1	0.9789	0.9790	+0.0001
Specificity	0.9969	0.9969	+0.0000
ROC AUC	0.9971	0.9969	-0.0002
MCC	0.9750	0.9751	+0.0001

5. Generalization Conclusion & Integrity Status

Status: STRONG GENERALIZATION. Performance remains virtually identical on unseen validation data ($|\Delta| = 0.01\% \leq 1.5\%$). All integrity checks passed (0 duplicate rows, 0 retraining calls, production artifacts untouched). The model achieves an Independent Validation Accuracy of **97.82%** and Weighted F1 of **0.9783**.